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UFW FIREWALL CONFIGURATION AND PORT VERIFICATION

PRACTICAL REPORT • TASK NO. 04

1. Objective

To configure UFW in Kali Linux, define inbound rules for selected TCP ports, identify the network interface and IP address, and perform a basic port-status check using Netcat.

2. Environment and Tools

Component	Details
Operating system	Kali Linux in VMware Workstation
Firewall	UFW
Network / port utilities	iproute2 (ip), Netcat (nc)
Interface / IPv4	eth0 — 10.122.110.11/24

3. Procedure and Commands

Command	Purpose
`sudo ufw default allow outgoing`	Allows outbound network connections by default.
`sudo ufw allow 22/tcp`	Adds an inbound allow rule for TCP port 22 (commonly SSH).
`sudo ufw deny 23/tcp`	Adds an inbound deny rule for TCP port 23 (commonly Telnet).
`sudo ufw enable`	Enables the UFW firewall.
`sudo ufw status verbose`	Displays firewall state, default policies and rules.
`ip -br addr`	Displays network interfaces and assigned IP addresses.
`sudo ufw status numbered`	Lists firewall rules with numbers.
`nc -lvp 23`	Starts a Netcat listener on TCP port 23.
`nc -vz -w 3 10.122.110.11 23`	Checks TCP connectivity to port 23 on the Kali host.

4. Observations and Results

UFW was shown as **ACTIVE**. The rules listed TCP/22 **ALLOW IN** and TCP/23 **DENY IN**, including IPv6 entries. The default policy was deny incoming and allow outgoing. The `ip -br addr` output identified **eth0** as UP with **10.122.110.11/24**.

A Netcat listener was started on TCP/23 and the connection check reported the port as **OPEN**. This confirms that the test connection reached a listener in the displayed setup; it does not prove that UFW blocked port 23. A local connection may not represent inbound traffic from another machine.

5. Conclusion and Verification Note

UFW was enabled and the intended rules were added. However, the blocking effect of the TCP/23 deny rule remains unverified because Netcat reported the port as open while a listener was running. Stop the listener and repeat the test from a separate authorized VM or host on the same lab network. The expected result is a blocked connection or timeout. Verify rules with `sudo ufw status numbered`.

6. Evidence Screenshots

Figure 1. UFW rules, Kali network address, and Netcat listener output.

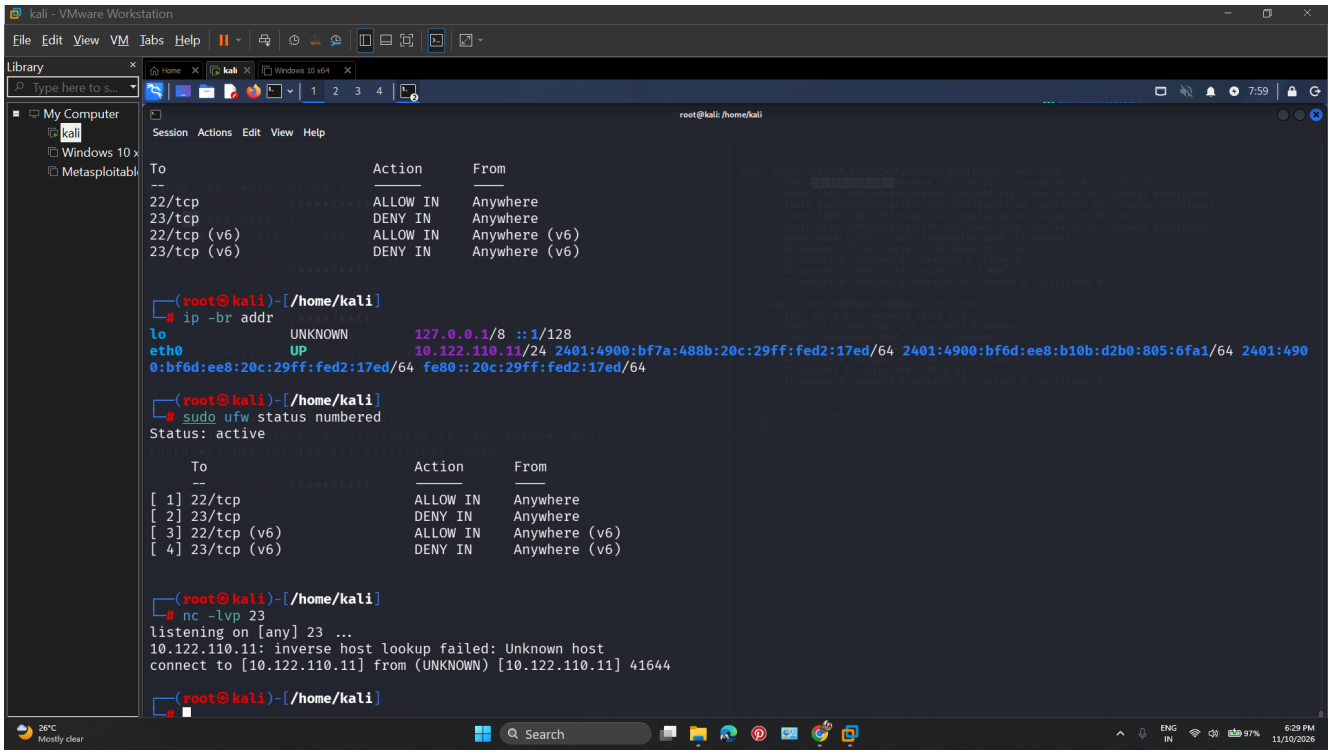


Figure 2. Netcat TCP port check against 10.122.110.11:23.

